

Snowflake Syllabus

Designed by- Sadhana Patil (Sr. Data Engineer)

Total Days: 25 Days (1.5 hr/day)

1. Snowflake Overview

- Data Types
- Datalake / Database / Datawarehouse
- OLTP & OLAP
- Normalization & Denormalization
- On-premise & Cloud

2. Snowflake Architecture

- What & why Snowflake
- Traditional Datawarehouse
- Cloud Warehouse
- Snowflake Architecture
 - Shared Disk Architecture
 - Shared Nothing Architecture
 - Multi-Cluster Shared Data Architecture
 - Snowflake Architecture Impact
- Advantages of Snowflake over On-premise
- Snowflake Features

3. Snowflake UI Overview

- Snowflake UI / Snowsight UI
- Snowflake Worksheets

4. Virtual Warehouse

- What is Virtual Warehouse
- Warehouse Size
- Scaling: Vertical & Horizontal
- Auto Suspend
- Auto Resume
- Hands-On
- Warehouse Billing
- Concurrency
- Query Queue
- Warehouse Best Practices

- Create Warehouse
- Resize Warehouse
- Suspend / Resume
- Monitor Credits

5. Database / Schemas / Tables / Stages

- Create Database & Schema
- Create Table
- Types of Tables and Their Properties
- Tables Types in Depth: Transient Tables and Dynamic Tables
- Stages
- Tasks

6. Caching

- What is Caching
 - Types of Caching

7. Time Travel

8. Fail-safe & Fail-safe Features

9. Zero Copy Clone

10. Micro-Partitions & Clustering

11. MV in Snowflake

12. Dynamic Tables

13. Stored Procedures

14. Full Refresh & Incremental Refresh

15. Introduction to Data Loading

- Loading Data into Snowflake
 - Load from Local Files
 - Load from Cloud Storage
 - Full Load vs Incremental Load
-

16. Snowflake Stages

- What is a Stage?
 - User Stage
 - Table Stage
 - Named Internal Stage
 - External Stage
 - Internal vs External Stage
-

17. File Formats

18. Stage Commands

- LIST
 - PUT
 - GET
 - REMOVE
-

19. COPY INTO

- What is COPY INTO?
 - Loading Files from Stage to Table
 - Basic COPY INTO Syntax
 - Loading Files
-

20. COPY INTO Options

- ON_ERROR
 - PURGE
 - FORCE
 - MATCH_BY_COLUMN_NAME
 - Validation Mode
-

21. Error Handling and Validation

- Handling Bad Record
 - ON_ERROR
 - VALIDATION_MODE
 - Error Files
 - Checking Rejected Records
 - Troubleshooting Load Failures
-

22. Loading from External Cloud Storage

- AWS S3
 - Azure Blob Storage
 - Google Cloud Storage
 - Storage Integration
 - External Stage
-

23. Snowpipe

- Why Snowpipe
 - Traditional COPY INTO vs Snowpipe
 - Continuous Data Loading
 - Auto Ingest
 - Event Notifications
 - Snowpipe Monitoring
-

24. Streams and Tasks

- Scheduling Snowflake Tasks
-

25. Query Performance Basics

26. Cost Optimization Basics

- Monitoring & Cost Optimization
-

27. Cortex CLI and Cortex AI

28. Streamlit Dashboards

DBT (Data Build Tool) – Syllabus

Designed by- Sadhana Patil (Sr. Data Engineer)

1. DBT Introduction

- What is DBT
- Why DBT Introduced
- Problems Solved by DBT
- Why Modern Data Teams Use DBT

2. Why DBT with Snowflake?

- Snowflake Performs Transformations
- DBT Manages SQL and Dependencies
- Separation of Storage and Compute
- Modular SQL Development
- Automated Documentation
- Testing Capabilities
- Version Control Integration

3. Features of DBT

- Modularity
- Version Control
- Documentation
- Testing
- Jinja Templating
- Dependency Management
- Reusability
- Environment Management

4. DBT Core vs DBT Cloud

- Architecture
- Features
- Advantages
- Limitations
- Which One to Use

5. Account Setup

5.1 Prerequisites

- Install Git
- Install Python
- Verify Python Version
- Install VS Code

5.2 Virtual Environment

- What is Virtual Environment?
- Why Virtual Environment?
- Create Virtual Environment
- Activate Virtual Environment

5.3 Install DBT

- Install dbt-snowflake
- Verify Installation

5.4 Initialize Project

- dbt init
- Project Structure
- Important Folders

5.5 Connect DBT with Snowflake

- profiles.yml
- dbt_project.yml

5.6 Validate Setup

- dbt debug
- Common Errors
- Troubleshooting
-

6. Understanding DBT Project Structure

- dbt_project.yml
 - Seeds
 - profiles.yml
 - Snapshots
 - models
 - Targets
 - macros
 - Logs
 - tests
-

7. Building the First DBT Project

7.1 Snowflake Setup

Create:

- Database
- Schemas
- RAW Tables
- Sample Data

7.2 Source Layer

- source.yml
- source()
- Why Sources

7.3 DBT Models

- Staging
- Intermediate
- Mart

7.4 Staging Layer

- First Model
- Naming Convention
- Data Cleaning
- Standardization

7.5 Intermediate Layer

- Business Logic
- Aggregation
- Calculations

7.6 Mart Layer

- Final Reporting Models
- Business Ready Data

7.7 Execute Models

- DBT Run
 - Model Dependency / DAG
 - Ref Function
 - Source Function
-

8. DBT Models

8.1 What is a Model?

- SQL File
- Transformation Logic
- Materialization

8.2 Model Types

- Views
- Table
- Incremental
- Ephemeral
- MV

8.3 Model Naming Standards

- int_
 - Stg
 - dim
 - fct_
 - mart_
-

9. Jinja Templating

9.1 Introduction

- What is Jinja
- Why Jinja

9.2 Variables in DBT

9.3 Loops in Jinja

9.4 Conditional Logic

9.5 Practical Examples: Used in Real Analytics Projects

- If
- For
- Set
- Comments

9.6 DBT Functions

10. Testing / Documentation

- Schema Tests
- Custom Tests

- Generic Tests
 - Freshness Tests
 - Document Generation
 - DBT Docs
-

11. Advanced DBT Features

- Macros
- Jinja Templating
- Variables
- Hooks
- Seeds
- Snapshots
- Packages
- Metric Layer Basics
- Incremental Load
- Implementing Snapshots